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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	"Hack into Google, No mistake"
Selected Domain	2
Selected SDG Track (1–17)	SDG 12.3 & SDG 8
Project / Solution Name	SOPBox: Smart Receiving Check for F&B SMEs
Name (Leader)	Lye Wei Lun
Email & Phone (Leader)	dellralllye@gmail.com & 018-960 6378
University Affiliation(s)	Tunku Abdul Rahman University of Management and Technology (TAR UMT)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and Medium Enterprises (SMEs) in the Food and Beverage (F&B) industry such as restaurants and cafes, often face challenges when receiving goods from suppliers. Many still depend on manual checking, where staff need to inspect products, count quantities, and record delivery details themselves. Although these tasks may seem simple, mistakes can easily happen, especially during busy working hours.

One common issue is receiving frozen or chilled products that are not within the required temperature range. If staff fail to notice this, the products may spoil or become unsafe for consumption, leading to food waste and financial losses. Besides that, products are often delivered in different packaging units, such as boxes, bundles, or packets. Incorrect quantity calculations can result in inaccurate inventory records and stock shortages.

Besides that, another challenge includes new staff may not be familiar with the proper receiving procedures. Without clear guidelines, different employees may handle deliveries differently. Managers may also find it difficult to monitor these issues when delivery records are incomplete or not properly maintained.

These problems can affect daily operations, increase business costs, and potentially compromise food safety. For small and medium F&B businesses with limited staff and resources, managing these processes consistently can be difficult.

Therefore, there is a need for a simple and affordable solution that helps F&B SMEs check deliveries more accurately, follow proper receiving procedures and keep reliable records while reducing unnecessary food waste.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SOPBox is an AI-powered receiving management system designed to help Small and Medium Food and Beverage (F&B) businesses manage their incoming deliveries more accurately and efficiently. It aims to reduce human errors, improve food safety, and make daily receiving tasks easier for both staff and managers.

The system allows staff to take a photo of a supplier's delivery order using their phone. An AI vision model processes the image and extracts important information, such as product names, quantities, and packaging units. Staff can then review and correct the extracted details before proceeding with the receiving check.

Once the information is confirmed, SOPBox automatically checks the delivery against predefined product rules, including acceptable temperature ranges, packaging conversions, and ordered quantities. If any issue is detected, the delivery is flagged for manager approval. Managers can decide whether to accept, reject, or return the delivery, while the system records the outcome for future reference.

Besides receiving checks, SOPBox provides a product catalogue, inventory tracking, purchase order management, and Standard Operating Procedure (SOP) cards to guide staff through proper receiving and handling procedures. Telegram notifications also help managers stay informed when their attention is needed.

To develop SOPBox, our team applies technical knowledge related to Generative AI APIs, database architecture, and cloud deployment introduced during the SDC Phase 1 workshops. We use Python and FastAPI for backend development, React and TypeScript for the user interface, PostgreSQL for data storage, RunPod for AI model inference, and Replit for application deployment.

By combining AI-powered data extraction with rule-based validation and human approval, SOPBox provides a practical and affordable solution for improving receiving operations in F&B SMEs.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

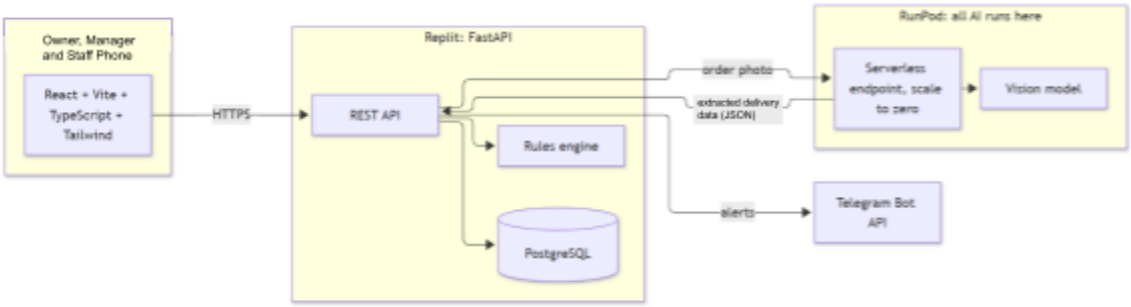
List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

SopBox uses a web-based architecture that connects AI-powered document extraction, automated validation, database management, and real-time notifications.

Component	Technology	Purpose
Frontend	React, TypeScript, Vite, Tailwind CSS	Mobile-friendly interface for staff, managers, and owners
Backend	Python, FastAPI	Handles business logic, API requests, and user authentication
AI Model	PaddleOCR-VL or Qwen3-VL	Extracts product information from delivery order images
AI Hosting	RunPod Serverless	Hosts the vision model for image processing
Database	Replit PostgreSQL	Stores accounts, products, SOPs, stock, deliveries, and audit records
Cloud Deployment	Replit	Hosts the web application and backend
Notifications	Telegram Bot API	Notifies managers of failed checks and staff of decisions
Security	scrypt, HttpOnly cookies, PostgreSQL RLS	Protects accounts and separates workspace data
Testing	pytest	Tests validation rules, API functionality, and security

Architecture Diagram :



Yes, our project utilises the official SDC sponsor-provided platforms, RunPod and Replit. RunPod is used to host and run the AI vision model for extracting information from delivery order images, while Replit is used to develop, host, and deploy the SOPBox web application, including its backend and database integration.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied a design-thinking approach by focusing on the daily challenges faced by small and medium F&B businesses, such as incorrect quantity calculations, missed temperature checks, and inconsistent receiving procedures.

To address these issues, we designed SOPBox to be simple and practical for everyday use. The system provides a mobile-friendly interface where staff can upload delivery orders, verify product details, and complete receiving checks without requiring advanced technical knowledge.

During development, we prioritised essential features, including user authentication, product management, receiving validation, manager approval, and inventory tracking. We also conducted automated testing to ensure the system functions correctly, validates delivery information accurately, and protects business data.

For deployment, SOPBox uses Replit to host the application and PostgreSQL to store business records securely. RunPod is planned for AI-powered document extraction, while manual entry remains available when AI processing is unavailable.

Before Phase 2, our team plans to test the system using sample delivery orders and realistic receiving scenarios. We will also gather user feedback and make improvements to ensure SOPBox is reliable, user-friendly, and suitable for real F&B operations.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 12.3 (Primary):** Reduce food waste and food losses by identifying delivery issues, such as incorrect temperatures, before products are accepted (United Nations, n.d.-b). **SDG 8.2 (Secondary):** Improve productivity through technology by reducing manual receiving tasks and helping staff work more efficiently (United Nations, n.d.-a).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	For our initial evaluation, we estimate that 5 out of 20 deliveries (25%) may contain temperature or quantity issues. Our team has completed automated system testing and tested SOPBox with a few sample delivery orders, where the information was successfully extracted and checked. We also estimate that manual receiving takes around 3–5 minutes per delivery, depending on the number of products. These figures will serve as our preliminary baseline for evaluating the system's performance.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**SDG 12.3:** Percentage of delivery issues correctly detected by SOPBox and estimated kilograms of potentially unsuitable food identified. **SDG 8.2:** Average receiving time per delivery (minutes) and percentage of time saved compared to manual checking.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**SDG 12.3:** We aim to achieve at least 90% accuracy in detecting delivery issues, such as incorrect temperatures and quantities, to help reduce the risk of accepting unsuitable products. **SDG 8.2:** We aim to reduce average receiving time by 20% compared to manual checking, making the process faster and more convenient for staff. These targets will build on our initial Phase 1 testing results.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React and React DOM	**code(frontend)**	**MIT**
Qwen3-VL-8B-Instruct (FP8)	**model**	**FP8**
vLLM	**model(runtime)**	**Apache2**
FastAPI	**code(Backend)**	**MIT**
Uvicorn	**code(Backend)**	**MIT**
Pydantic	**code(Middleware)**	**MIT**
Httpx	**code(Middleware)**	**BSD-3**
psycopg and psycopg-pool	**code (Backend)**	**LGLP-3**
PostgreSQL	**code(Database)**	**PostgreSQL License**
Lucide icons	**code(frontend)**	**ISC**
SimpleWebAuthn Browser	**code(frontend)**	**MIT**
Fontsource	**code(frontend)**	**SIL Open Font License**
Playwritght	**test(frontend)**	**Apache 2.0**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 24/10/2026